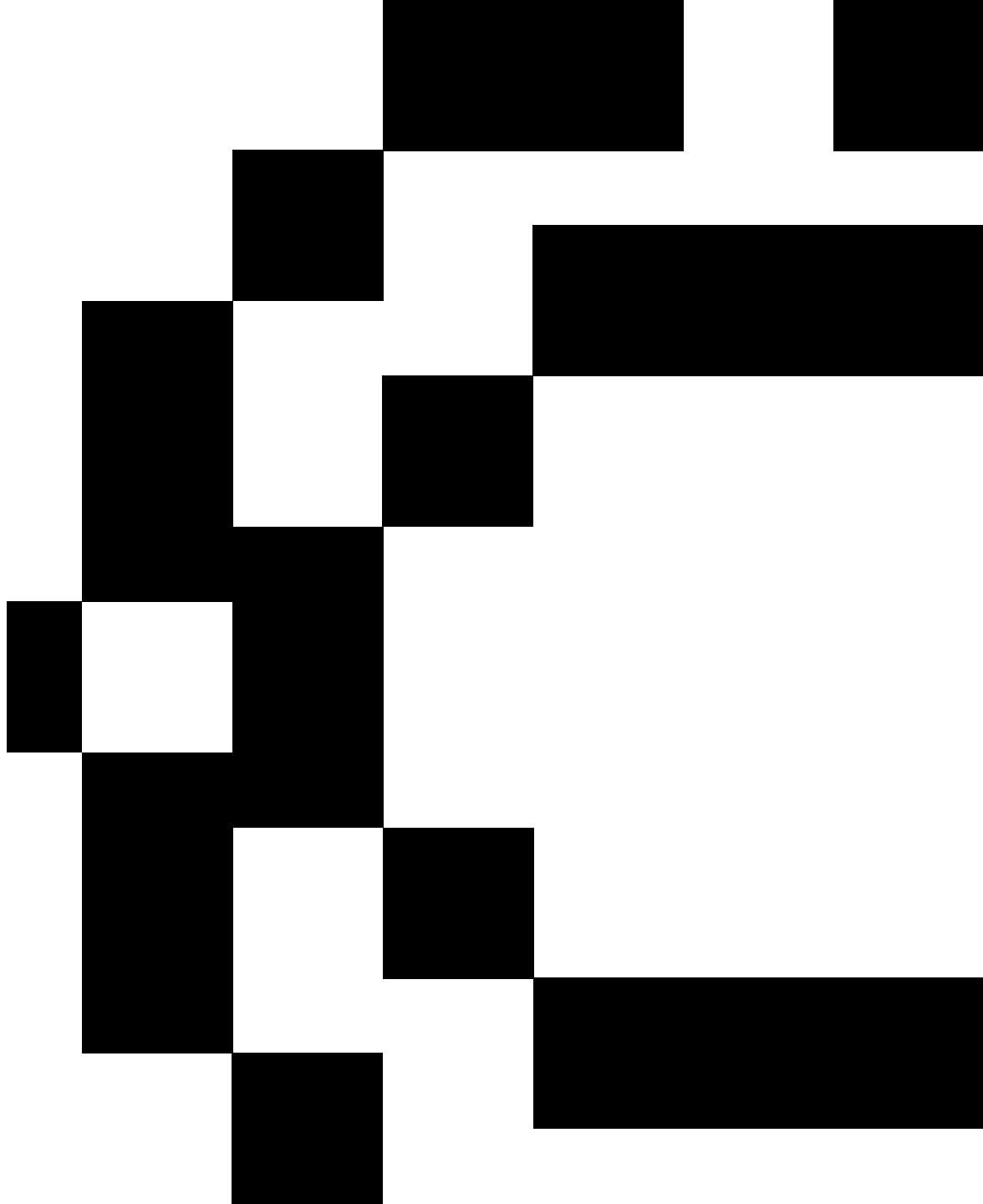
NOVA IMS INFORMATION MANAGEMENT SCHOOL

Computation I / Introduction to Programming

Practicals Week 04 - Input Handling

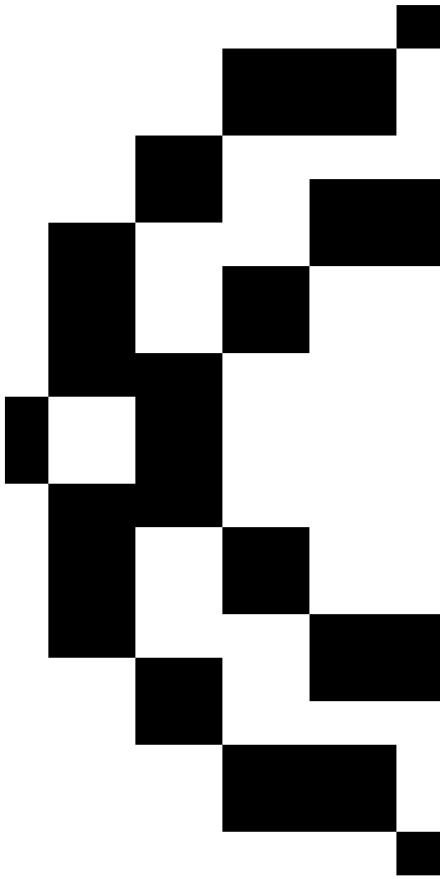


Exercise 1

Ask the user for their favourite film and print it back inside a sentence.

Example

- `Input:  movie = "Inception"`
- `Output: Your favourite movie is Inception`



Exercise 1 — Solution

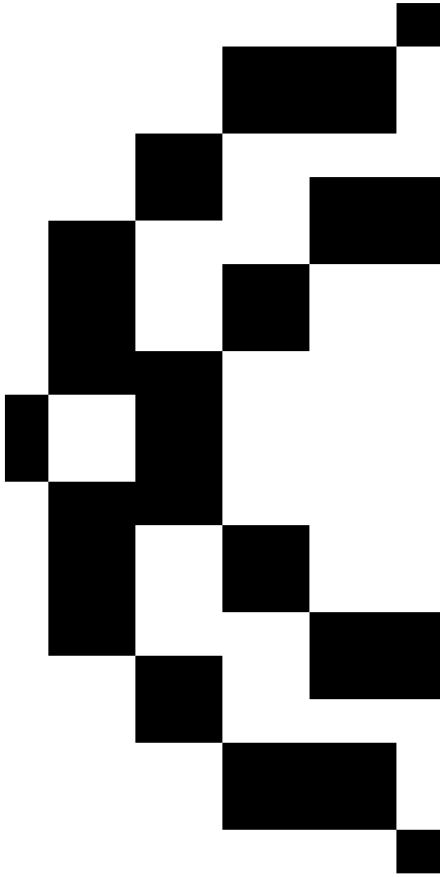
```
movie = input("What is your favourite movie? ")
print(f"Your favourite movie is {movie}")
```

Exercise 2

Ask the user for a word and a number, then print the word repeated that number of times, with no separator between repetitions.

Example

- `Input:  word = "ha", number = 3`
- `Output: hahaha`



Exercise 2 — Solution

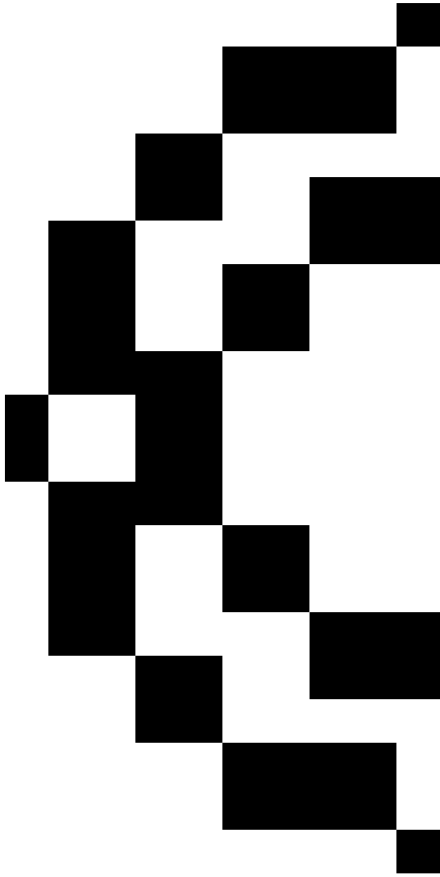
```
word = input("Insert a word: ")
number = int(input("Insert a number: "))
print(word * number)
```

Exercise 3

Ask the user for a number of hours and a number of minutes, then calculate and print the total number of seconds.

Example

- `Input:  hours = 1, minutes = 30`
- `Output: Your total is 5400 seconds.`



Exercise 3 — Solution

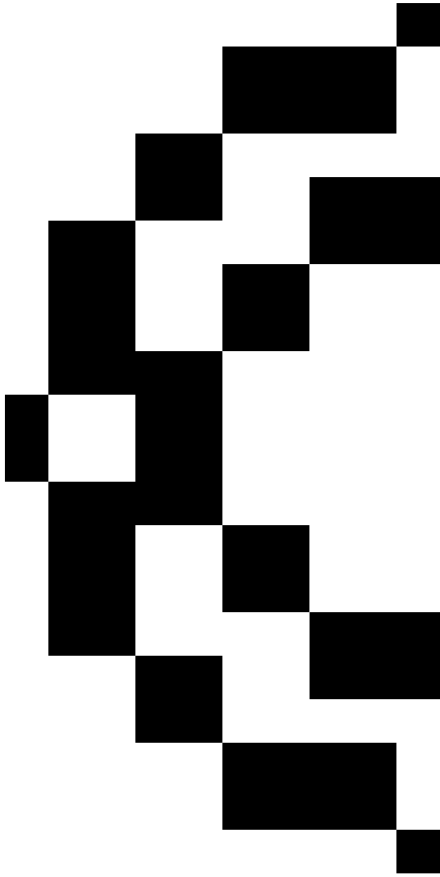
```
hours = int(input("How many hours? "))
minutes = int(input("How many minutes? "))
total_minutes = hours * 60 + minutes
total_seconds = total_minutes * 60
print(f"Your total is {total_seconds} seconds.")
```

Exercise 4

Ask the user for a sentence and an index, then print the character at that index within the sentence.

Example

- `Input:  sentence = "Hello world", index = 6`
- `Output: w`



Exercise 4 — Solution

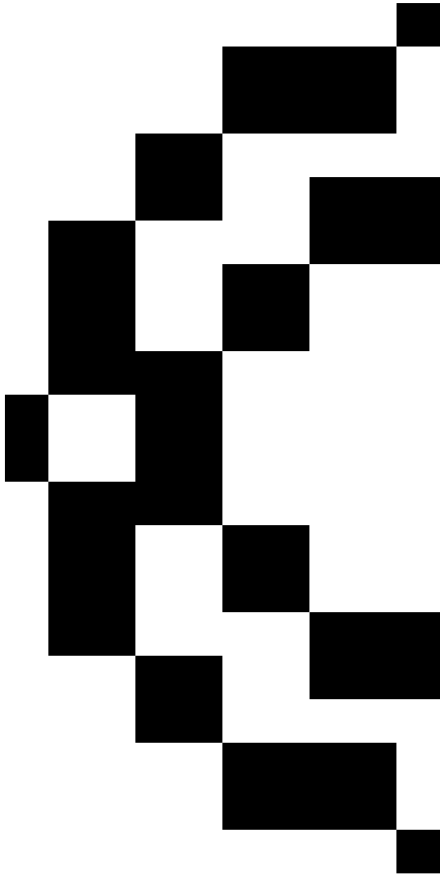
```
sentence = input("Insert a sentence: ")
index = int(input("Insert an index: "))
print(sentence[index])
```

Exercise 5

Ask the user for a number, then print `True` or `False` for each of the following questions: is it positive, is it even, is it divisible by 3, is it divisible by 5, and does it fall in the interval [-10, 10]?

Example

```
Input:  number = 12
Output: Is it positive? True
        Is it even? True
        Is it divisible by 3? True
        Is it divisible by 5? False
        Is it in the range [-10,10]? False
```



Exercise 5 — Solution

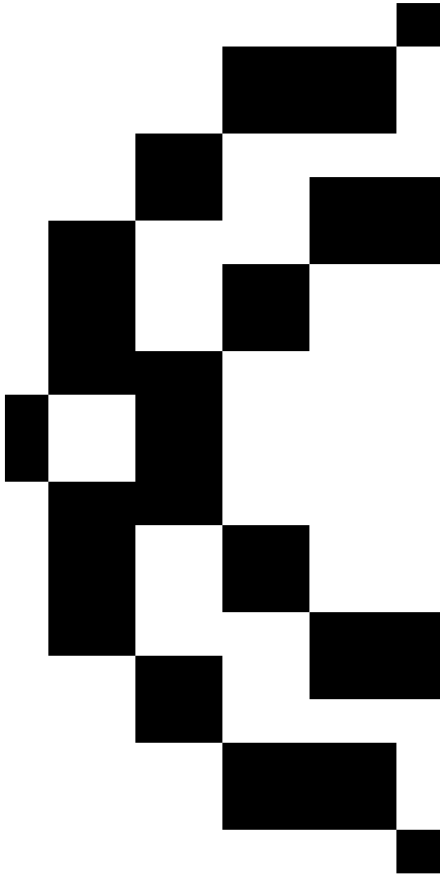
```
number = float(input("Insert a number: "))
print(f"Is it positive? {number > 0}")
print(f"Is it even? {number % 2 == 0}")
print(f"Is it divisible by 3? {number % 3 == 0}")
print(f"Is it divisible by 5? {number % 5 == 0}")
print(f"Is it in the range [-10,10]? {-10 <= number <= 10}")
```

Exercise 6

Create a list containing five numbers of your choosing. Print the first, third, and last elements using positive indexes, then repeat using negative indexes. Change the middle value to 42. Finally, ask the user for a number and add it to the end of the list, then print the whole list.

Example

```
Input:  lst = [3, 10, 2, 5, 20], new number = 7
Output: 3 2 20
        3 2 20
        [3, 10, 42, 5, 20, 7]
```



Exercise 6 — Solution

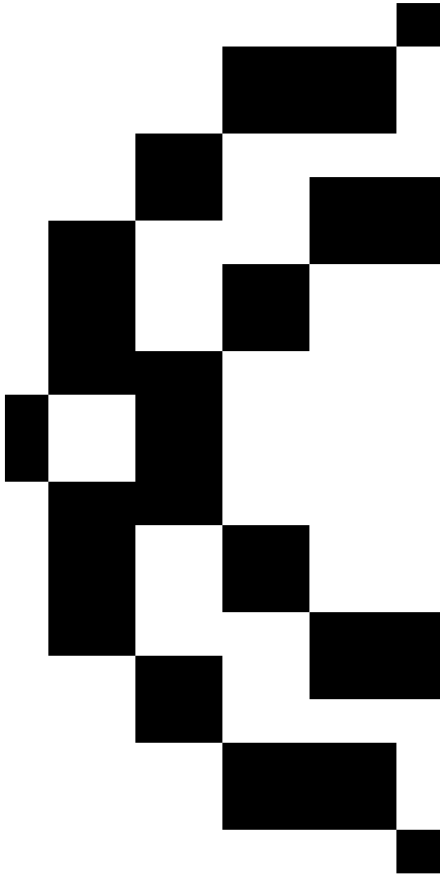
```
lst = [3, 10, 2, 5, 20]
print(lst[0], lst[2], lst[4])
print(lst[-5], lst[-3], lst[-1])
lst[2] = 42
new_number = int(input("Please enter a number: "))
lst.append(new_number)
print(lst)
```

Exercise 7

Ask the user to type two halves of a sentence as `half1` and `half2`. Print the complete sentence formed by `half1`, followed by a comma and a space, followed by `half2`.

Example

- `Input:  half1 = "I am going to the beach", half2 = "but I have to finish my lessons first!"`
- `Output: I am going to the beach, but I have to finish my lessons first!`



Exercise 7 — Solution

```
half1 = input("Enter the first part of a phrase: ")
half2 = input("Enter the second part of that phrase: ")
print(f"{half1}, {half2}")
```

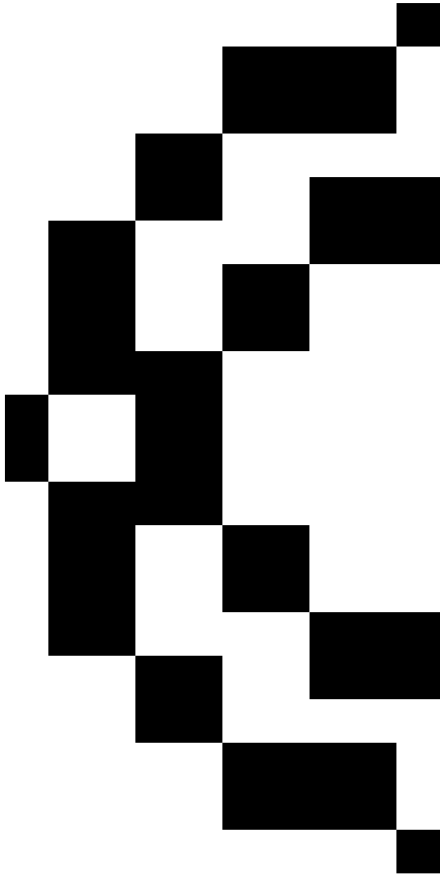
Exercise 8

Ask the user to type their favourite colour. Print `True` if the colour appears in the list `red, white, black, green, orange, purple`, and `False` otherwise. The check should not depend on how the user capitalises their answer.

Examples

- `Input:  colour = "Red"`
- `Output: True`

- `Input:  colour = "Blue"`
- `Output: False`



Exercise 8 — Solution

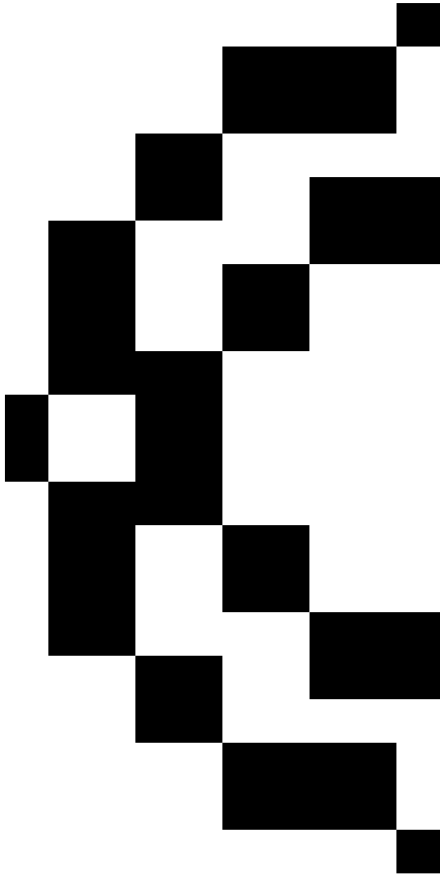
```
colour = input("Insert your favourite colour: ")
colour_string = "red, white, black, green, orange, purple"
print(colour.lower() in colour_string)
```

Exercise 9

Ask the user to enter a date in `DD/MM/YYYY` format. Print the day, month, and year separately, in a sentence.

Example

- `Input:  dt = "25/12/2026"`
- `Output: The day is 25, the month is 12, the year is 2026.`



Exercise 9 — Solution

```
dt = input("Enter a date in the format DD/MM/YYYY: ")
day = dt[0:2]
month = dt[3:5]
year = dt[-4:]
print(f"The day is {day}, the month is {month}, the year is {year}.")
```

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Thank you

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